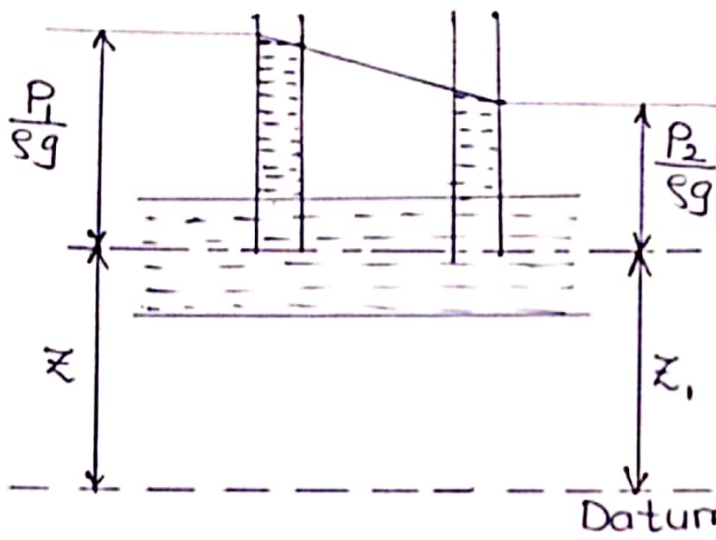
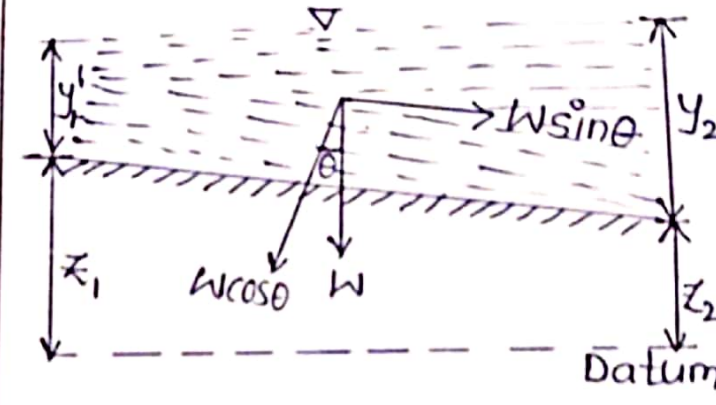


Open Channel Flow

①

* Difference between pipe flow & OCF.

Pipe Flow	OCF
	
① In pipe flow, fluid flow thr^u pressure difference. ② $Re = \frac{\text{Inertia force}}{\text{Viscous force}}$	① In OCF, fluid flow through Gravity ② $Fr = \sqrt{\frac{\text{Inertia Force}}{\text{Gravitational force}}}$

* Types of flow :-

① Steady and Unsteady flow :-

If flow parameters are constant w.r. to time, so flow is considered as a steady flow otherwise, flow is called unsteady flow.

② Uniform and Non-uniform flow :-

If flow parameters are constant w.r. to space so flow is called uniform flow otherwise flow is called Non-uniform flow.

③ Rotational and Irrotational flow:-

If fluid particles rotate about its own axis so flow is called rotational flow. Otherwise flow is considered as irrotational flow.

④ Vortex flow and Non-vortex flow:-

If all the fluid particles are rotate w.r. to common axis so is called vortex flow.

Types of vortex flow → ① Free Vortex
② Force Vortex.

⑤ Laminar, Transitional & Turbulent flow:-

For Pipe flow:- $Re < 2000$ (Laminar flow)
 $2000 \leq Re \leq 4000$ (Transitional flow)
 $Re > 4000$ (Turbulent flow)

⑥ Sub-critical, Critical and Super critical flow:-

$$Fr = \frac{\sqrt{\text{Inertia force}}}{\sqrt{\text{Gravitational force}}}$$

$$Fr = \frac{V}{\sqrt{gD}} \quad \text{where, } D = \text{Hydraulic Depth}$$

$Fr < 1$ (Sub-critical flow)

$Fr = 1$ (critical flow)

$Fr > 1$ (super-critical flow)

* Types of Channel -

① Artificial & Natural channel :-

Artificial channel are man-made channels such as canals & Natural channels are river.

② Prismatic and Non-prismatic channel :-

Cross-section area is constant along it's length is called prismatic channel. Otherwise it is non-prismatic channel.

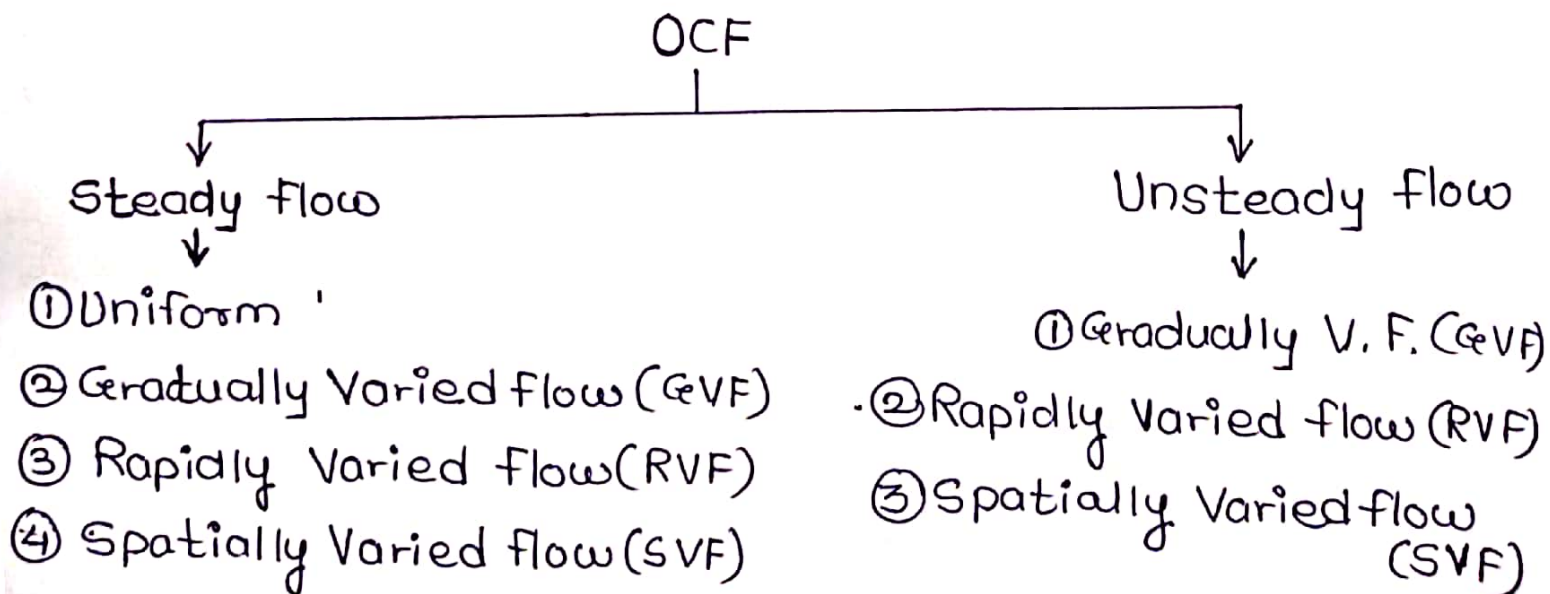
③ Rigid boundary And Mobile boundary channel :-

boundary is not deform w.r.to time is called rigid boundary otherwise it is mobile boundary channel.

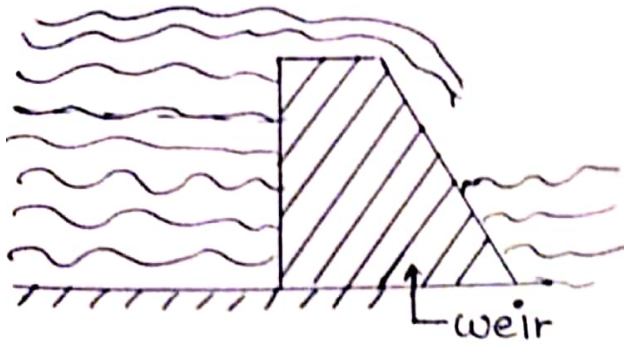
Note -

In OCF, we assumed that channel is Artificial, prismatic & Rigid boundary channel.

* classification of Steady and unsteady flow :-

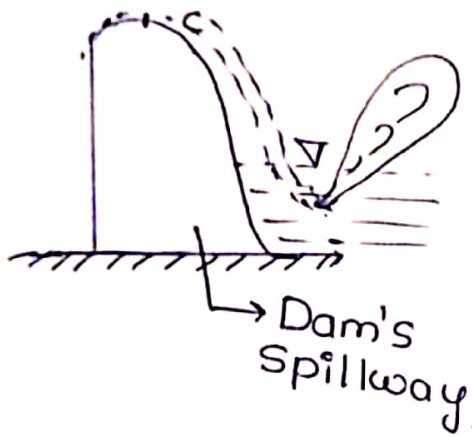


① Gradually Varied Flow :-



Depth of flow is gradually increases at upstream of weir.

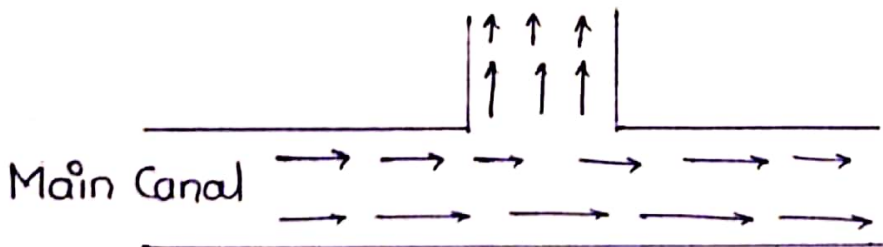
② Rapidly Varied Flow :-



when water falls over spillway hydraulic jump is form & Level of water rapidly increases, is called rapidly varied flow.

③ Spatially Varied flow :-

Some amount of water added into a system or some amount of water subtract into a system so flow is considered as Spatially varied flow.
eg. Canal Network.



Difference Between Dam, Weir & Barrage.

Dam:- Dam is hydraulic structure which construct across the river or natural stream.

- The main function is stored the water.
- Dam is part of storage scheme.
- It is always be construct on impervious foundation. (i.e. rock)
- Dam is solid structure.
- Flow is not permitted over Dam structure.
- Transportation is ~~not~~ permitted.

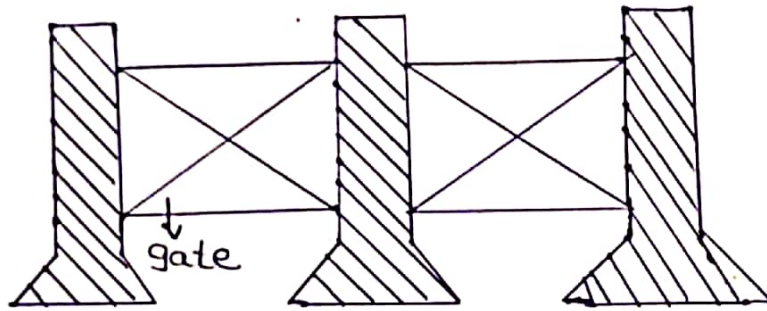
Weir:- Weir is hydraulic structure which is construct across river or Natural stream.

- The main function is to divert the water.
- It is part of divergence scheme.
- weir is a solid structure
- Major ponding of water is done by solid obstruction & Minor ponding of water is done by gatted obstruction.
- During flood, weir is submerged structure.
- It is construct on perious foundation.
- Transportation is not permitted over weir.
- Cost of construction is low ~~over~~ as compare to Barrage.
- Value of afflux is high.

Barrage :-

⑥

- ① Major ponding of water is done by gated obstruction & Minor ponding of water is done by solid obstruction.
- ② It is uneconomical structure.
- ③ Main function is to divert the water.
- ④ It is a part of divergence scheme.
- ⑤ Barrage is not submerged structure.
- ⑥ Transportation is allowed over Barrage.
- ⑦ Barrage is a uneconomical structure.
- ⑧ Afflux is low.



Note:-

Sometime Dam is also used in Divergence scheme but In this case dam is called as pick-up weir.

Pick-up weir → it is construct like a dam & act like a weir.

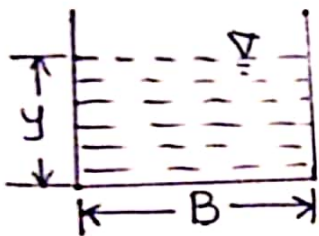
Some Important Parameter :-

Channel:- channel is a medium or path using which water is transmit through one section to another section.

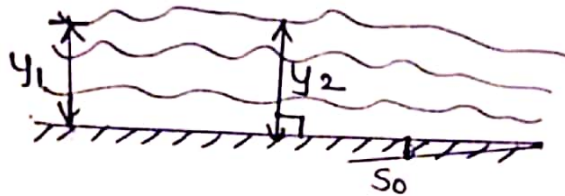
Depth of flow & Depth of flow Section :-

Vertical depth w.r. to channel bed is called Depth of flow (y_1)

Vertical depth w.r. to $\perp$ lar to channel bed is called Depth of flow section (y_2)



c/s view.

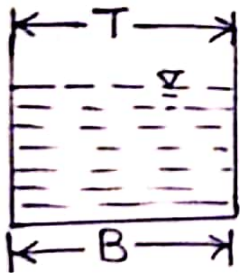


S_0 = Longitudinal Slope.

when S_0 is very small $\therefore y_1 = y_2$

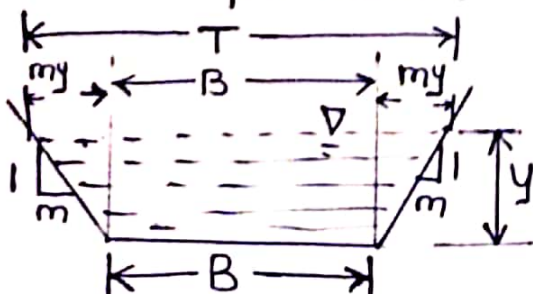
Top Width (T) :-

For Rectangular channel, width = Top width



$$\therefore B = T$$

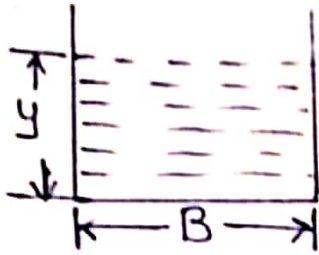
For Trapezoidal channel,



$$\therefore T = B + 2my$$

Wetted Area & Wetted Perimeter :-

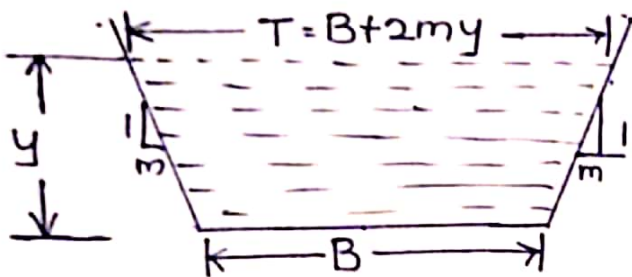
For Rectangular Channel :-



$$\text{Wetted Area (a)} = B \cdot y$$

$$\text{Wetted Perimeter (P)} = B + 2y$$

For Trapezoidal Channel :-



$$\text{Wetted Area } A = \frac{1}{2}(B + B + 2my)y$$

$$= \frac{1}{2} [2B + 2my]y$$

$$\boxed{A = \frac{1}{2} (B + my)y}$$

$$\text{Wetted Perimeter} = P =$$

$$\boxed{P = B + 2y\sqrt{1 + m^2}}$$

Hydraulic Depth :- (D)

Ratio of wetted Area to Top width

$$D = \frac{\text{Wetted Area}}{\text{Top width}} = \frac{A}{T}$$

$$\boxed{D = \frac{A}{T}}$$

Rectangular Channel :-

$$B = T$$

$$\therefore D = \frac{B \cdot y}{B} = y$$

$$\boxed{\therefore D = y}$$

$$Fr = \frac{V}{\sqrt{gD}} \quad \because D = y$$

$$\boxed{\therefore Fr = \frac{V}{\sqrt{gy}}}$$

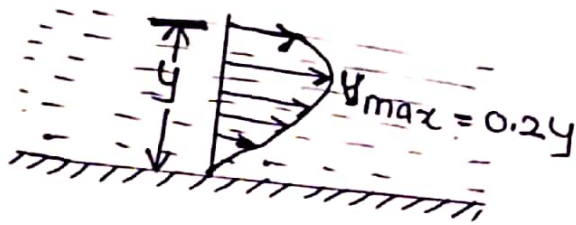
Hydraulic Mean Radius (R):-

Ratio of wetted Area to wetted perimeter.

$$R = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}}$$

$$R = \frac{A}{P}$$

* Velocity Distribution in OCF:-



$$V_{\max} = 0.2y$$

$$V_{\min} = 0.8y$$

$$V_{\text{avg}} = 0.6y \quad \{ \text{Shallow River} \}$$

$$V_{\text{avg}} = \frac{V_{0.2y} + V_{0.8y}}{2} \quad \{ \text{Deep River} \}$$

$$\frac{B}{y} < 5 \quad - \text{Deep \& Narrow}$$

$$\frac{B}{y} > 5 \quad - \text{shallow \& wide}$$

B = width of channel

y = Depth of flow.

$$\frac{B}{y} = \text{Aspect Ratio}$$

* Imp. Eqⁿ for calculation of velocity & Discharge (10)

① Chezy's Equation :-

$$V = C\sqrt{RS}$$

$$50 \leq C \leq 70$$

where S = Longitudinal slope
 R = Hydraulic mean depth
 C = chezy's constant

$$C = \sqrt{\frac{8g}{f}}$$

f = friction coefficient
 g = gravity acceleration,

Following Eqⁿ are used to find value of C :-

① Kutter's Equation :-

$$C = \frac{23 + \frac{0.00155}{S} + \frac{1}{n}}{1 + \left(23 + \frac{0.00155}{S}\right) \cdot \frac{n}{\sqrt{R}}}$$

n = Roughness Coefficient

{value of n depends on channel bed material}

S = Longitudinal Slope

R = Hydraulic Mean Depth.

② Bazin's Equation :-

$$C = \frac{157.6}{1.81 + \frac{m}{\sqrt{R}}}$$

m = Bazin's Coefficient

{value of m depends on
Channel bed material}

③ Manning's Equation :-

$$V = \frac{1}{N} R^{2/3} S^{1/2}$$

N = Manning's Roughness Coeff

S = Longitudinal slope

R = Hydraulic depth

$$\boxed{N = \frac{(K)^{1/6}}{24}} \quad \text{OR}$$

$$\boxed{N = \frac{(D_{50})^{1/6}}{24}}$$

K = conveyance factor.

D_{50} - 50% particles are finer than this size.

$$V = C \sqrt{RS} = C \cdot R^{1/2} \cdot S^{1/2} \quad \text{--- (1)}$$

$$V = \frac{1}{N} R^{2/3} S^{1/2} \quad \text{--- (2)}$$

equate equation (1) & (2)

$$C \cdot R^{1/2} S^{1/2} = \frac{1}{N} R^{2/3} S^{1/2}$$

$$\boxed{C = \frac{1}{N} R^{1/6}}$$

Discharge Calculation: -

$$\begin{aligned} Q &= A \cdot V \\ &= A \frac{1}{N} R^{2/3} S^{1/2} \end{aligned}$$

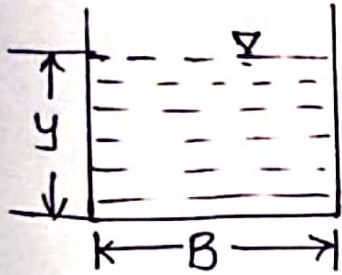
$$\boxed{Q = \frac{1}{N} \cdot A \cdot R^{2/3} S^{1/2}}$$

$K = AR^{2/3}$ = conveyance factor.

Most Economical Channel Section :-

Channel is said to be most economical when its construction cost is minimum for a given discharge.

① Most Economical Rectangular Channel :-



$$P = B + 2y \quad \text{--- ①}$$

$$A = B \cdot y \quad \therefore B = \frac{A}{y} \quad \text{--- put value of } B \text{ in eqn ①}$$

$$\therefore P = \frac{A}{y} + 2y \quad \text{--- ②}$$

For minimum perimeter, -

$$\frac{dP}{dy} = 0.$$

$$-\frac{A}{y^2} + 2 = 0$$

$$-\frac{B \cdot y}{y^2} + 2 = 0 \quad (\because A = By)$$

$$-\frac{B}{y} + 2 = 0. \quad \therefore B = 2y \quad \boxed{\therefore y = \frac{B}{2}}$$

(i) Hydraulic Mean depth :-

$$R = \frac{A}{P} = \frac{B \cdot y}{B + 2y}$$

$$= \frac{(2y)y}{2y + 2y} = \frac{2y^2}{4y} \quad (\because B = 2y)$$

$$\boxed{\therefore R = \frac{y}{2}}$$

ii) Wetted Area (A) = iii) Top Width (T) -

(3)

$$A = By$$

$$\therefore A = 2y \cdot y$$

$$\boxed{A = 2y^2}$$

$$T = B$$

$$\boxed{T = 2y} \quad (\because B = 2y)$$

iv) Wetted Perimeter:-

$$P = B + 2y$$

$$= 2y + 2y = \underline{4y}$$

$$\boxed{P = 4y}$$

v) Hydraulic Depth (D) -

$$D = \frac{A}{T} = \frac{A}{B}$$

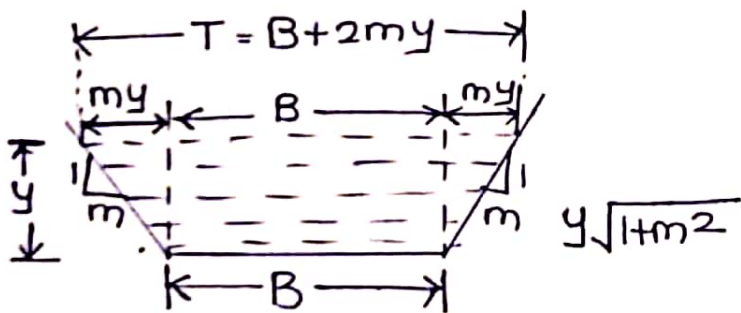
$$= \frac{2y^2}{2y} = y$$

$$\boxed{D = y}$$

vi) Froude No. -

$$Fr = \frac{V}{\sqrt{gD}} = \frac{V}{\sqrt{gy}}$$

Most Economical Trapezoidal Channel Section:-



Wetted perimeter = P

$$\boxed{P = B + 2y\sqrt{1+m^2}} \quad \text{--- (1)}$$

$$A = \frac{1}{2} (B + (B + 2my)) y$$

$$A = \frac{1}{2} [2B + 2my] \cdot y$$

$$A = (B + my) \cdot y$$

$$\therefore B = \frac{A}{y} - my$$

Put value of B in eqn (1)

$$\boxed{P = \frac{A}{y} - my + 2y\sqrt{1+m^2}}$$

For most economical channel section:-

$$\frac{dP}{dy} = 0$$

$$2\sqrt{1+m^2} = \frac{B+2my}{y} \quad \text{--- (1)}$$

$$2y\sqrt{1+m^2} = B+2my \quad \text{--- (2)}$$

i) Hydraulic Mean depth (R):-

$$R = \frac{A}{P}$$

$$= \frac{(B+my)y}{B+2y\sqrt{1+m^2}}$$

$$= \frac{(B+my)y}{B+B+2my}$$

$$= \frac{(B+my)y}{2B+2my}$$

$$= \frac{(B+my)y}{(B+my)2}$$

$$\boxed{R = \frac{y}{2}}$$

iii) ~~Width of channel~~:-

Wetted Area (A) =

$$A = (B+my)y$$

$$= \left(\frac{2y}{\sqrt{3}} + \frac{y}{\sqrt{3}} \right) y$$

$$A = \frac{3y}{\sqrt{3}} y = \frac{3y^2}{\sqrt{3}}$$

$$= \frac{\sqrt{3}\sqrt{3} y^2}{\sqrt{3}}$$

$$\boxed{\therefore A = \sqrt{3} y^2}$$

For Best slope:-

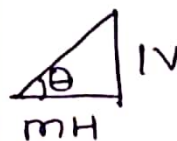
$$\frac{dP}{dm} = 0$$

$$\frac{d}{dm} \left(\frac{A}{y} - my + 2y\sqrt{1+m^2} \right) = 0$$

$$\boxed{m = 1/\sqrt{3}}$$

ii) ~~Wetted Area (A) =~~

$$A = (\cancel{B+my})y$$



$$\tan \theta = \frac{1}{m}$$

$$\tan \theta = \sqrt{3}$$

$$\therefore \theta = \tan^{-1}(\sqrt{3})$$

$$\boxed{\therefore \theta = 60^\circ}$$

Put value of m in eqn (1)

$$2y\sqrt{1+m^2} = B+2my \quad \text{--- (1)}$$

$$2y\sqrt{1+\frac{1}{3}} = B+2\left(\frac{1}{\sqrt{3}}\right)y$$

$$2y\left(\frac{2}{\sqrt{3}}\right) = B+\frac{2}{\sqrt{3}}y$$

$$\boxed{\frac{2y}{\sqrt{3}} = B}$$

Length of side slope

$$= y\sqrt{1+m^2}$$

$$= y\sqrt{1+\frac{1}{3}}$$

$$= y\left(\frac{2}{\sqrt{3}}\right)$$

$$\boxed{\therefore \text{Length of side slope} = \frac{2y}{\sqrt{3}}}$$

iv) Wetted Perimeter:-

$$P = B + 2y\sqrt{1+m^2}$$

$$= \frac{2y}{\sqrt{3}} + 2y\sqrt{1+\frac{1}{3}}$$

$$= \frac{2y}{\sqrt{3}} + \frac{4y}{\sqrt{3}} = \frac{6y}{\sqrt{3}}$$

$$\therefore P = \frac{6y}{\sqrt{3}}$$

v) Top Width:- (5)

$$T = B + 2my$$

$$= \frac{2y}{\sqrt{3}} + 2 \times \frac{1}{\sqrt{3}} y$$

$$= \frac{2y}{\sqrt{3}} + \frac{2y}{\sqrt{3}}$$

$$\therefore T = \frac{4y}{\sqrt{3}}$$

vi) Hydraulic Depth (D):-

$$D = \frac{A}{T} = \frac{\frac{\sqrt{3}y^2}{4}}{\frac{4y}{\sqrt{3}}}$$

$$D = \frac{3}{4} y$$

vii) Froude No. -

$$Fr = \frac{V}{\sqrt{gD}} = \frac{V}{\sqrt{g\left(\frac{3}{4}\right)y}}$$

$$\therefore Fr = \frac{V}{\sqrt{g\left(\frac{3}{4}\right)y}}$$

Most Economical Triangular channel Section:-

$$A = \left(\frac{1}{2} \times y + my\right)^2$$

$$\therefore A = my^2$$

$$A = \tan\theta \cdot y^2 \quad \text{--- (1)}$$

$$P = 2y\sqrt{1+m^2}$$

$$= 2y\sqrt{1+\tan^2\theta}$$

$$\therefore P = 2y \cdot \sec\theta$$

For most economical channel section -

$$\frac{dP}{d\theta} = 0$$

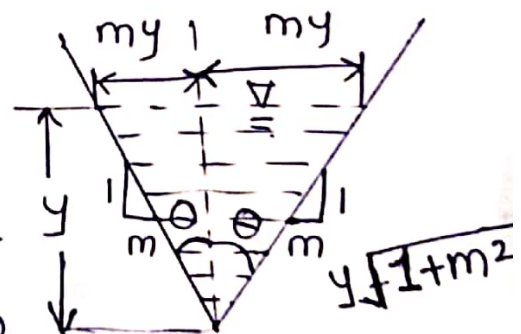
$$\tan\theta = \frac{my}{y}$$

$$\therefore \tan\theta = m$$

$$\frac{d}{d\theta} (2y \cdot \sec\theta) = 0$$

$$\therefore \theta = 45^\circ$$

$$\therefore A -$$



$$m = \tan\theta$$

$$= \tan 45^\circ$$

$$\therefore m = 1$$

i) Wetted Area

$$A = my^2$$

$$\therefore A = y^2$$

ii) Wetted Perimeter

$$P = 2y\sqrt{1+m^2}$$

$$\therefore P = 2y\sqrt{1+1}$$

$$\therefore P = 2\sqrt{2}y$$

(6)

iii) Hydraulic Mean Radius (R)

$$R = \frac{A}{P} = \frac{y^2}{2\sqrt{2}y}$$

$$\therefore R = \frac{y}{2\sqrt{2}}$$

v) Hydraulic Depth (D)

$$D = \frac{A}{T} = \frac{y^2}{2y} = \frac{y}{2}$$

$$\therefore D = \frac{y}{2}$$

iv) Top width

$$T = 2my$$

$$T = 2y$$

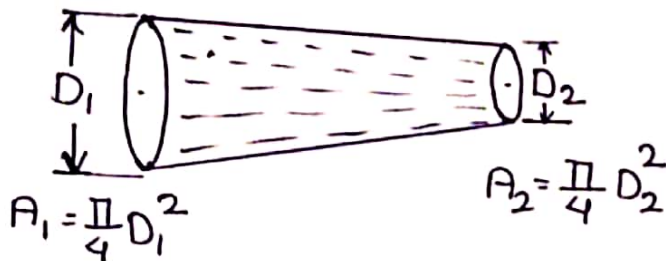
vi) Froude No.

$$Fr = \frac{V}{\sqrt{gD}}$$

$$Fr = \frac{V}{\sqrt{g(\frac{y}{2})}}$$

Continuity Equation:-

* This concept based on conservation of mass.

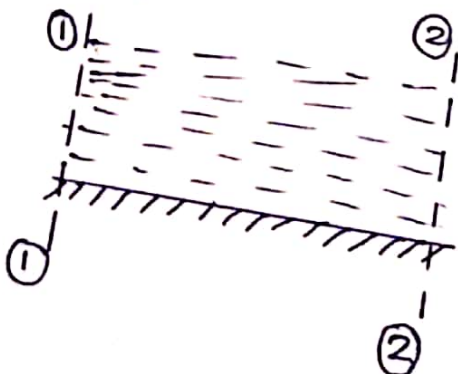


$$Q_1 = Q_2$$

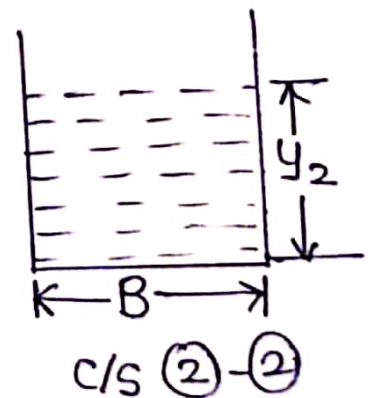
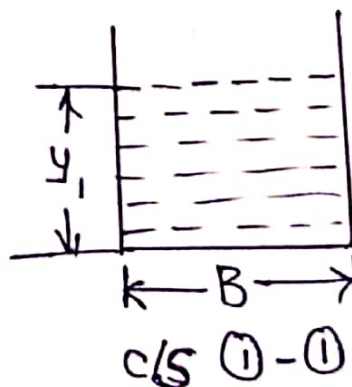
$$A_1 V_1 = A_2 V_2$$

* Steady flow:-

Rectangular Open channel:-



L-Section



$$A_1 = B.y_1 \quad A_2 = B.y_2$$

$$(\text{Discharge})_{1-1} = (\text{Discharge})_{2-2}$$

$$Q_1 = Q_2$$

$$A_1 V_1 = A_2 V_2$$

$$\therefore B y_1 v_1 = B y_2 v_2$$

$$\therefore y_1 v_1 = y_2 v_2$$

$$y \uparrow v \downarrow \rightarrow F_r < 1 \text{ (Subcritical flow)}$$

$y \downarrow v \uparrow \rightarrow F_r > 1$ (supercritical flow)

For Rectangular Channel - (17)

$$Q = A \cdot V,$$

$$Q = B \cdot y \cdot v$$

$$\frac{Q}{B} = 4. V$$

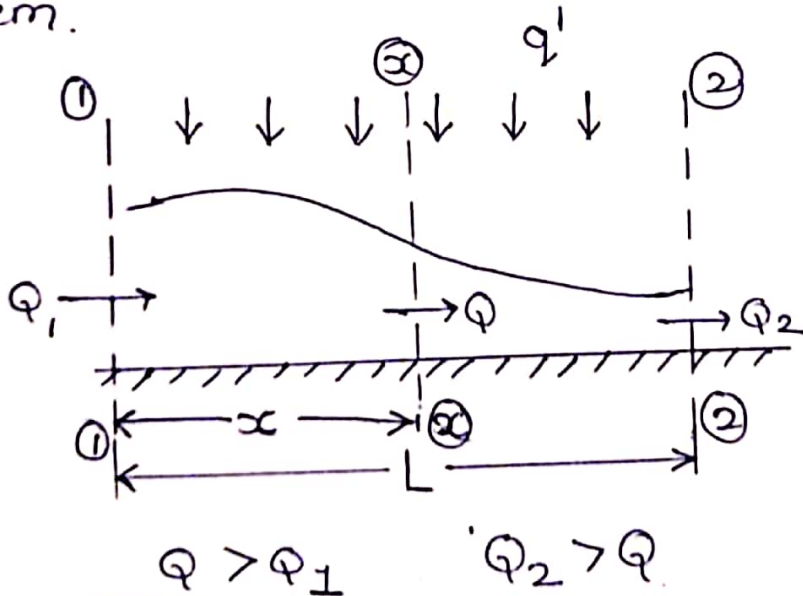
$$\frac{Q}{B} = q = \text{Discharge per unit width.}$$

$$\therefore q = y.v$$

$$\text{unit} = \frac{\text{m}^3}{\frac{\text{s}}{\text{m}}}$$

Spatially Varied Steady flow:-

Some amount of water added into a system OR some amount of water subtract into a system.



$$Q = Q_1 + q_1' x$$

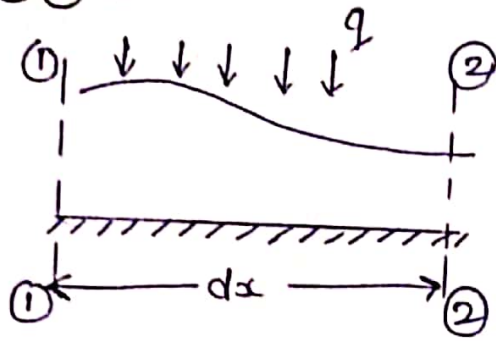
$$Q_2 = Q + q'(L-x)$$

$$\therefore \phi_2 > \phi_1$$

$$Q_2 = Q_1 + q'(L)$$

Unsteady Flow :-

case ① Some amount of water added into a system

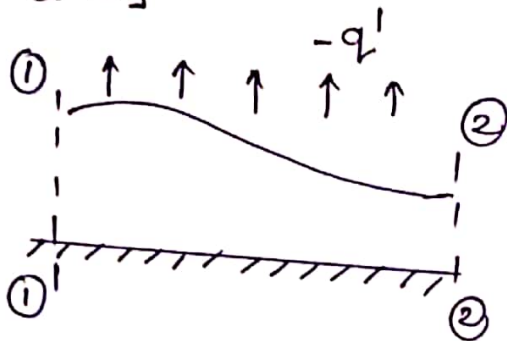


$$dQ = (Q_2 - Q_1)$$

$$dQ = +q \cdot dx$$

$$\frac{dQ}{dx} = +q$$

case ② Some amount of water subtract into a system.



$$dQ = -q' \cdot dx$$

$$\frac{dQ}{dx} = -q'$$

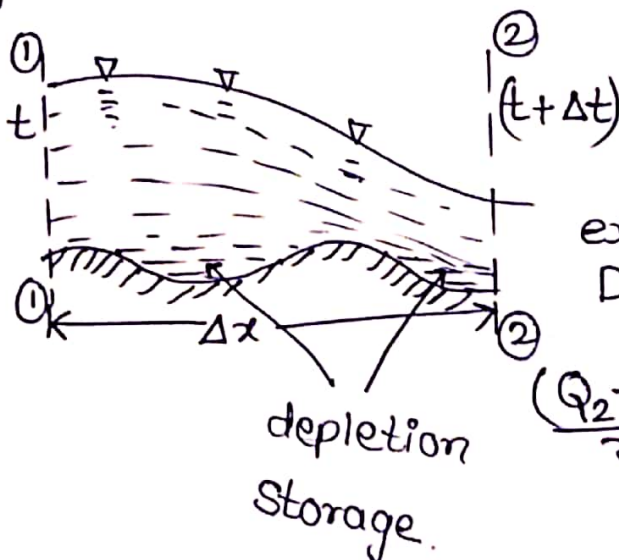
In General -

$$\frac{dQ}{dx} = \pm q$$

+ → added
- → Subtract.

Unsteady Flow :-

case ③



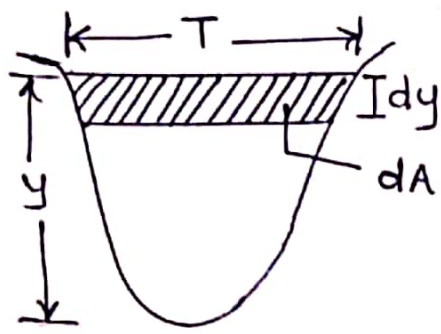
$$(Q_2 > Q_1) \text{ or } (Q_1 > Q_2)$$

due to depletion storage.

excess Rate of Discharge = - Rate of Depletion Storage

$$\frac{(Q_2 - Q_1)}{\Delta x} \Delta x \cdot \Delta t = - \frac{\partial}{\partial t} (A \cdot \Delta x) \cdot \Delta t$$

①



Consider a channel c/s. of any shape.
consider a strip of depth of dy
& width T .

Area of strip $dA = T dy$

$$\boxed{\frac{dA}{dy} = T}$$

From eqⁿ ①,

$$\frac{(\phi_2 - \phi_1)}{\partial x} \cdot \Delta x \cdot \Delta t = - \frac{\partial}{\partial t} (A \cdot \Delta x) \cdot \Delta t$$

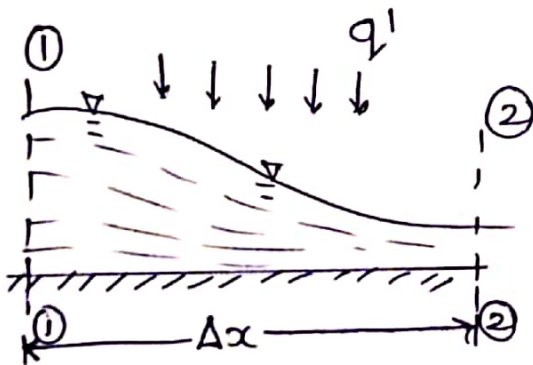
$$\frac{\partial \phi}{\partial x} \Delta x \cdot \Delta t = - \frac{\partial A}{\partial y} \cdot \frac{\partial y}{\partial t} \cdot \Delta x \cdot \Delta t$$

$$\frac{\partial \phi}{\partial x} = - T \frac{\partial y}{\partial t}$$

$$\boxed{\therefore \frac{\partial \phi}{\partial x} + T \frac{\partial y}{\partial t} = 0}$$

This is continuity eqⁿ for unsteady flow.

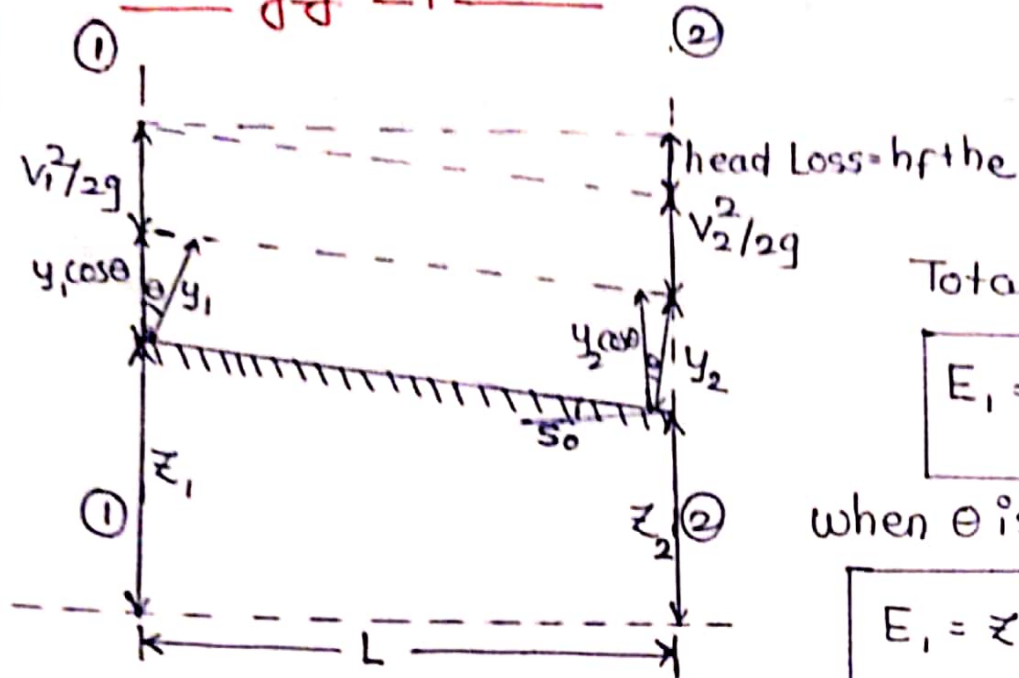
* Spatially Varied Unsteady flow:-



$$\frac{\partial \phi}{\partial x} + T \frac{\partial y}{\partial t} = \pm q'$$

$+q'$ = Some amount of water added into a system

$-q'$ = Some amount of water subtract into a system.

Energy Equation :-Total energy at sectⁿ ①-①

$$E_1 = z_1 + y_1 \cos \theta + \frac{V_1^2}{2g}$$

when θ is very small.

$$E_1 = z_1 + y_1 + \frac{V_1^2}{2g}$$

Total energy

$$= z_1 + y_1 + \frac{V_1^2}{2g}$$

Total energy at ②-②

$$E_2 = z_2 + y_2 \cos \theta + \frac{V_2^2}{2g}$$

$$E_2 = z_2 + y_2 + \frac{V_2^2}{2g}$$

Specific energy :- It is a total energy per unit wt. above channel bottom or channel bed.

at section ①-①, $E_1 = y_1 + \frac{V_1^2}{2g}$

at section ②-②, $E_2 = y_2 + \frac{V_2^2}{2g}$

specific energy eqⁿ.

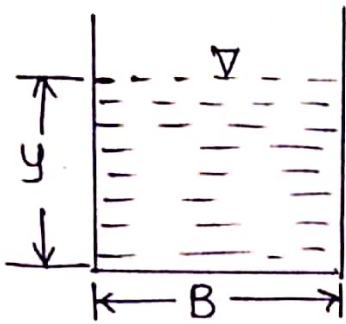
$$E = y + \frac{V^2}{2g}$$

$$Q = A \cdot V \quad V = \frac{Q}{A}$$

$$\therefore E = y + \frac{Q^2}{2g A^2}$$

In case of Rectangular channel:-

(2)



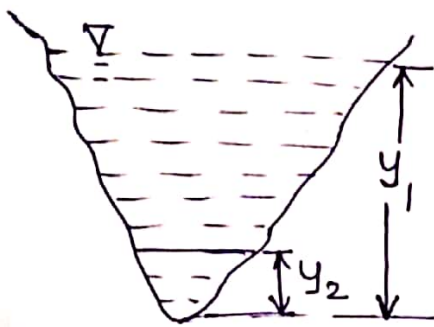
$$E = y + \frac{V^2}{2g}$$

$$E = y + \frac{Q^2}{2gA^2} \quad \left(\because Q = AV, V = \frac{Q}{A} \right)$$

$$E = y + \frac{Q^2 B^2}{2g(B^2 y^2)} \quad \left(q = \frac{Q}{B}, Q = qB \right)$$

$$E = y + \frac{q^2}{2gy^2}$$

Specific energy curve:-

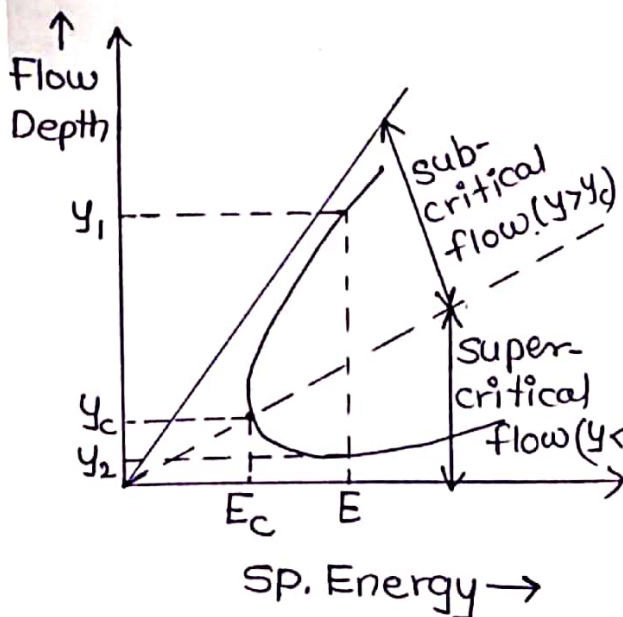


* If depth is increasing, so sp. energy linearly increases.

* If depth is decreasing, sp. energy ~~parabola~~ Hyperbolic type decreases.

$y \uparrow \rightarrow E \uparrow$ (Linearly)

$y \downarrow \rightarrow E \downarrow$ (Hyperbolically)



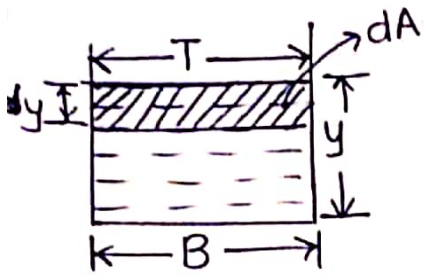
y_1 & y_2 are alternative depth, where sp. energy is same.

$\rightarrow y_c$ is called critical depth where sp. energy is min^m . (critical sp. energy)

Note:- If sp. energy is less than critical sp. energy, so flow is not possible.

Criteria for minimum specific energy:-

(22)



$$E = y + \frac{V^2}{2g}$$

$$E = y + \frac{Q^2}{2gA^2}$$

$$dA = T \cdot dy$$

For minimum sp. energy:-

$$\boxed{\frac{dA}{dy} = T}$$

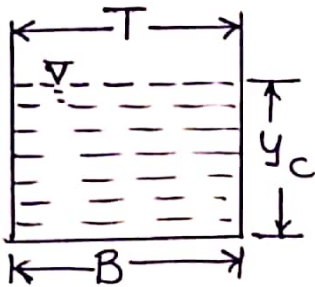
$$\therefore \frac{dE}{dy} = 0$$

$$\therefore 1 - \frac{2Q^2}{2gA^3} \cdot \frac{dA}{dy} = 0$$

$$\therefore 1 - \frac{Q^2 T}{gA^3} = 0$$

$$\boxed{\therefore \frac{Q^2 T}{gA^3} = 1}$$

For Rectangular Channel:-



$$\boxed{B = T}$$

→ At critical depth, sp. energy is min^m.

For minimum sp. energy:-

$$\frac{Q^2 T}{gA^3} = 1$$

$$\frac{q^2 B^2 \cdot B}{g B^3 y_c^3} = 1 \quad \left\{ \begin{array}{l} \frac{Q}{B} = q \therefore Q = q \cdot B \\ T = B \end{array} \right\}$$

$$\frac{q^2}{g y_c^3} = 1 \quad \therefore y_c^3 = \frac{q^2}{g}$$

$$\boxed{\therefore y_c = \left(\frac{q^2}{g} \right)^{1/3}}$$

∴ Critical sp. energy -

$$E_c = y_c + \frac{Q^2}{2gA_c^2}$$

$$E_c = y_c + \frac{q^2 B^2}{2g B^2 y_c^2}$$

$$\boxed{\therefore E_c = y_c + \frac{q^2}{2g y_c^2}}$$

Special Case:-

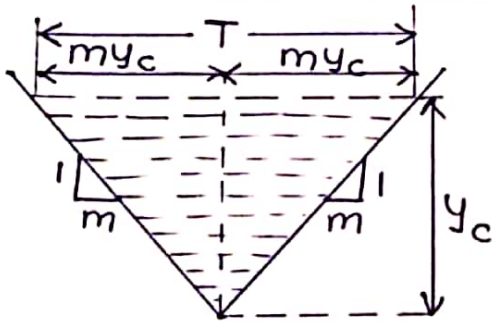
(23)

For rectangular channel, $y_c^3 = \frac{q^2}{g}$, $E_c = y_c + \frac{q^2}{2gy_c^2}$ — (1)

put $\frac{q^2}{g} = y_c^3$ in eqⁿ (1) $\therefore E_c = y_c + \frac{y_c^3}{2y_c^2}$

$$E_c = \frac{3}{2} y_c$$

For Triangular channel:-



$$\text{Top width (T)} = 2my_c$$

For minimum sp. energy.

$$\frac{Q^2 T}{g A^3} = 1$$

put value of A & T

$$A = 2 \times \frac{1}{2} \times my_c^2$$

$$A = my_c^2$$

$$\frac{Q^2 (2my_c)}{g m^3 y_c^6} = 1$$

$$\frac{2Q^2}{g m^2 y_c^5} = 1$$

$$\frac{2Q^2}{g m^2} = y_c^5$$

$$\frac{2Q^2}{g} = m^2 y_c^5$$

$$\therefore y_c = \left(\frac{2Q^2}{g m^2} \right)^{1/5}$$

1v:mH

For Critical sp energy

$$E_c = y_c + \frac{v^2}{2g}$$

$$E_c = y_c + \frac{Q^2}{2g A_c^2}$$

$$= y_c + \frac{g m^2 y_c^5}{2 \times 2g \cdot A_c^2}$$

$$= y_c + \frac{m^2 y_c^5}{4m^2 y_c^4}$$

$$= y_c + \frac{y_c}{4}$$

$$\therefore E_c = \frac{5y_c}{4}$$

Summary :-

(24)

① For Rectangular channel

$$y_c = \left(\frac{q^2}{g} \right)^{1/3}$$

$$E_c = \frac{3}{2} y_c$$

② For Triangular channel

$$y_c = \left(\frac{2Q^2}{gm^2} \right)^{1/5}$$

$$E_c = \frac{5}{4} y_c$$

Criteria for Maximum Discharge :-

$$E = y + \frac{V^2}{2g}$$

$$E = y + \frac{Q^2}{2gA^2}$$

$$\therefore Q = \sqrt{(E-y)2gA^2}$$

For Max^m Discharge,

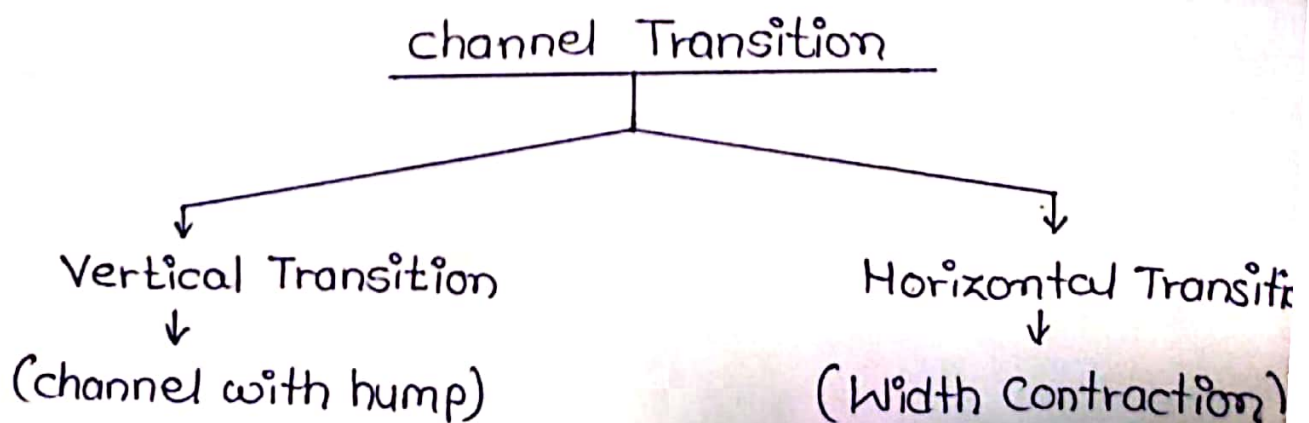
$$\frac{dQ}{dy} = 0$$

$$\boxed{\frac{Q^2 T}{gA^3} = 1}$$

Note:- Maximum discharge is obtained for given specific energy at critical depth.

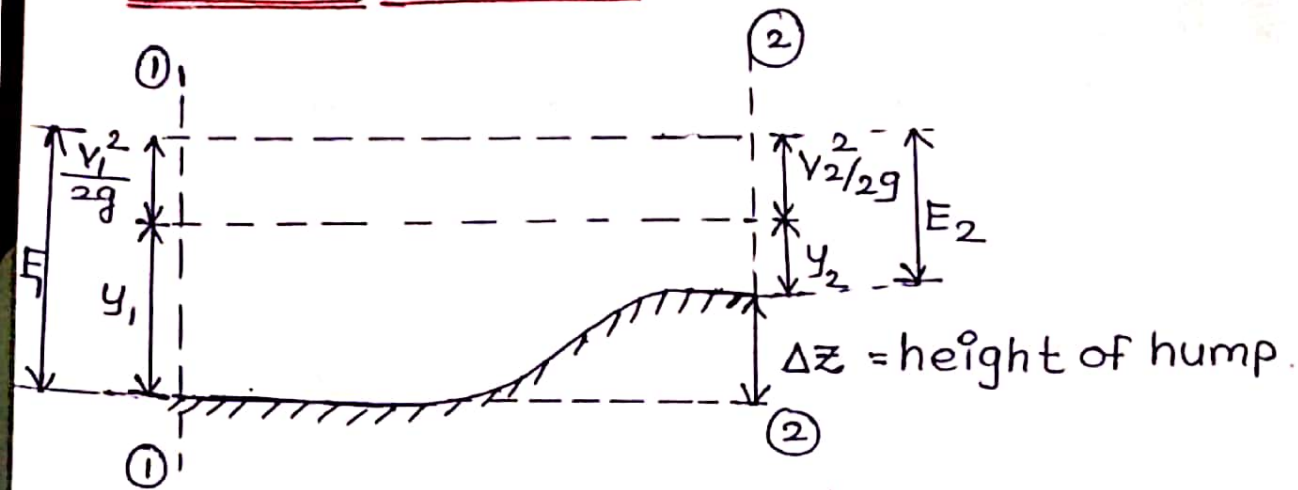
Channel Transition :-

→ A Transition is a portion of a channel with varying c/s which connect one uniform flow channel to another uniform flow channel.



① Vertical Transition :-

25



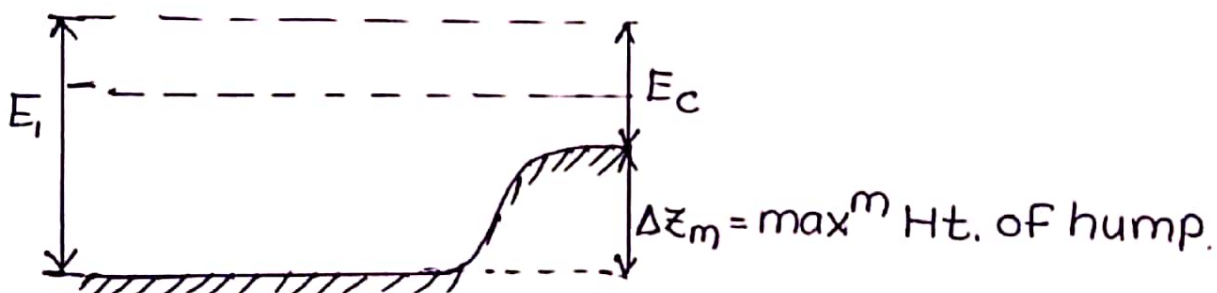
$$\Delta z = E_1 - E_2$$

At section ①-①

$$E_1 = y_1 + \frac{v_1^2}{2g}$$

At section ②-②

$$E_2 = y_2 + \frac{v_2^2}{2g}$$



$$\Delta z_m = E_1 - E_c$$

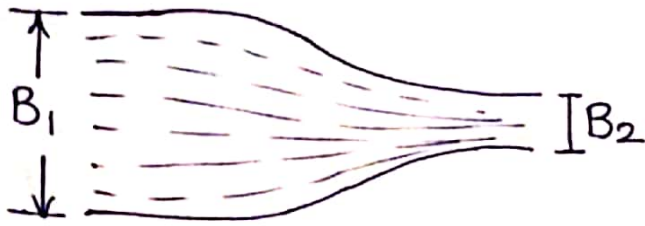
$$\Delta z \leq \Delta z_m \text{ OR } E_c \leq E_2$$

$\therefore$ Flow is possible

$$\Delta z > \Delta z_m \text{ OR } E_2 < E_c$$

$\therefore$ Flow is not possible, hence choking condition will occur

② Horizontal Contraction (Width Contraction) (26)



Top View

$$E_1 = y_1 + \frac{V_1^2}{2g}$$

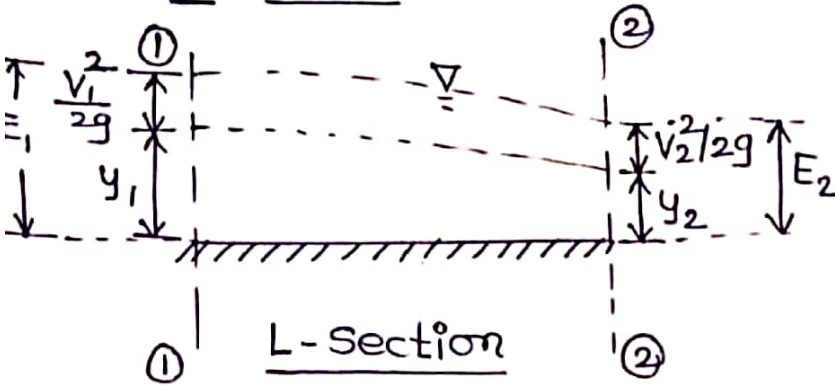
$$= y_1 + \frac{Q_1^2}{2gA_1^2}$$

$$= y_1 + \frac{Q_1^2}{2g \cdot B_1 y_1^2}$$

$$= y_1 + \frac{q_1^2 B_1^2}{2g \cdot B_1 y_1} \quad (Q_1 = q_1 B_1)$$

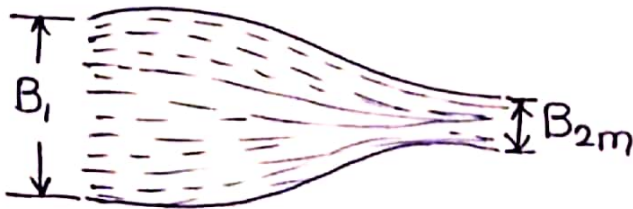
$$E_1 = y_1 + \frac{q_1^2 B_1}{2g y_1}$$

$$E_2 = y_2 + \frac{q_2^2 B_2}{2g y_2}$$



L-Section

B_{2m} = maximum width contraction

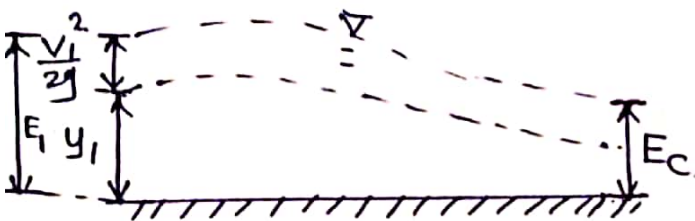


$$B_2 \geq B_{2m} \quad \text{OR} \quad E_2 \geq E_c$$

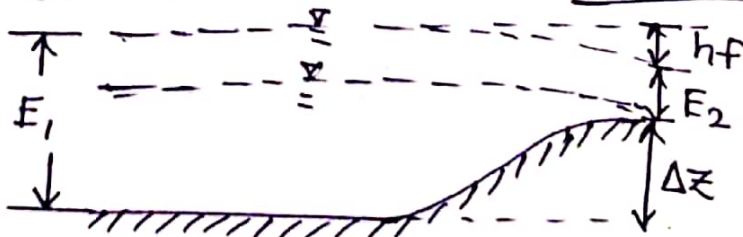
{Flow is possible}

$$B_2 < B_{2m} \quad \text{OR} \quad E_2 < E_c$$

{flow is not possible & choking condition will occur}



If channel is not frictionless :-



$$E_1 = E_2 + h_f + \Delta z$$

$$\therefore \Delta z = (E_1 - E_2) - h_f$$

Non-Uniform flow

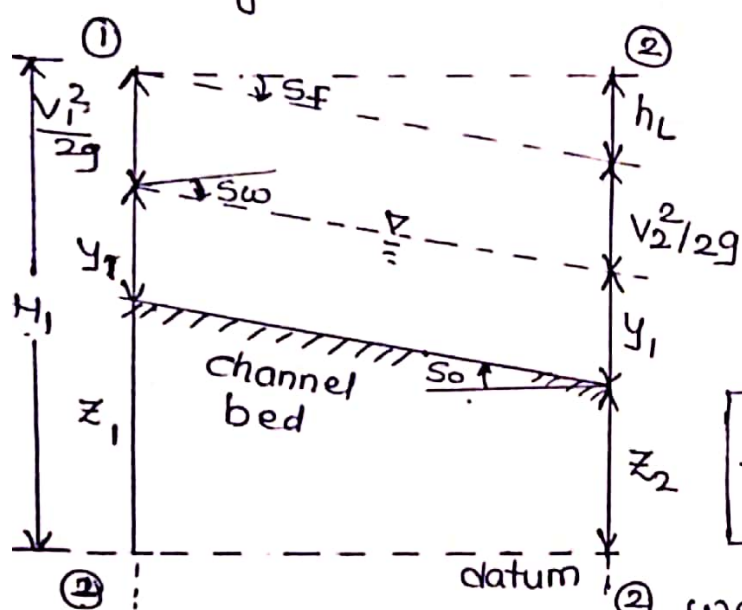
→ Flow parameters are not constant w.r. to space.

① Gradually Varied Non-uniform flow:-

→ Gradually Varied Non-uniform flow, flow depth vary gradually.

→ In G.V.N.U.F., Losses are neglected.

→ Only one loss is considered due to boundary friction.



Total energy/Total head-

$$H = z + y + \frac{V^2}{2g} \quad \text{--- ①}$$

Diff. eqⁿ ① w.r. to x .

$$\frac{dH}{dx} = \frac{dz}{dx} + \frac{dy}{dx} + \frac{d}{dx} \left(\frac{V^2}{2g} \right) \quad \text{--- ②}$$

② we know that,

$$\frac{dH}{dx} \rightarrow \text{slope of total energy line} \quad \therefore \frac{dH}{dx} = -S_f$$

$$\frac{dz}{dx} \rightarrow \text{slope of channel Bottom} \quad \frac{dz}{dx} = -S_o$$

put value in eqⁿ ②,

$$-S_f = -S_o + \frac{dy}{dx} + \frac{d}{dx} \left(\frac{V^2}{2g} \right)$$

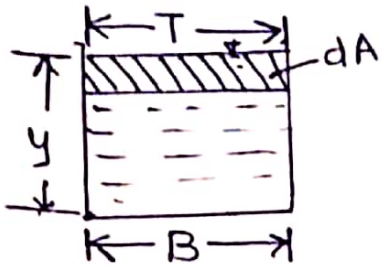
$$\frac{dy}{dx} = \frac{S_o - S_f}{1 - \frac{Q^2 T}{g A^3}}$$

$$\frac{dy}{dx} = \text{slope of water Surface} \quad \frac{dy}{dx} = S_w$$

$$S_w = \frac{S_o - S_f}{1 - \frac{Q^2 T}{g A^3}} \quad \text{--- ③}$$

$$F_r = \frac{V}{\sqrt{gD}} \quad [\text{for Rectangular channel } (y = D)]$$

$$\therefore F_r = \frac{V}{\sqrt{gy}} \quad \text{--- (4)}$$



$$dA = T dy$$

$$\therefore dy = \frac{dA}{T} \quad \therefore y = \frac{A}{T} \quad \text{Put in eqn (4)}$$

$$\therefore F_r = \frac{V}{\sqrt{g\left(\frac{A}{T}\right)}}$$

$$\therefore F_r = \frac{Q\sqrt{T}}{A\sqrt{gA}} \quad \left\{ \begin{array}{l} \because Q = AV \\ V = \frac{Q}{A} \end{array} \right\}$$

Taking sq. on both sides,

$$F_r^2 = \frac{Q^2 T}{A^2 \cdot g \cdot A} = \frac{Q^2 T}{g A^3} \quad \text{--- put in eqn (3)}$$

$$\therefore S_w = \frac{S_o - S_f}{1 - F_r^2}$$

Liquid Surface profile in OCF

→ Surface flow profile is depends on -

- i) Bottom slope of the channel
- ii) Depth of flow (y)
- iii) critical depth (y_c)
- iv) Normal depth of flow (y_n)

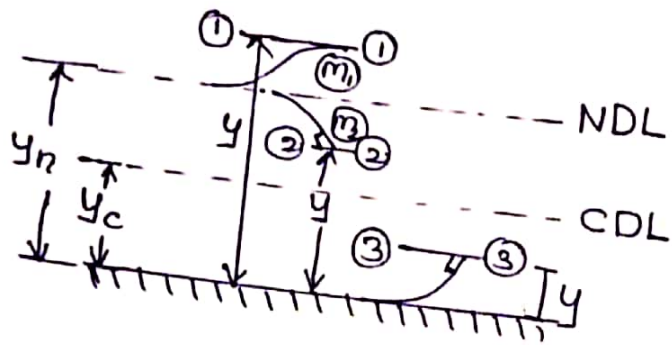
$y_n > y_c$ - subcritical flow

$y_n < y_c$ - Supercritical flow.

$y_n = y_c$ - critical flow.

i) Bottom Slope of channel:-

① Mild Slope:- $y_n > y_c$ - subcritical flow.



NDL - Normal Depth Line.
CDL - critical Depth Line.

At section ①-①,
(M₁ flow profile)

$$y > y_n > y_c$$

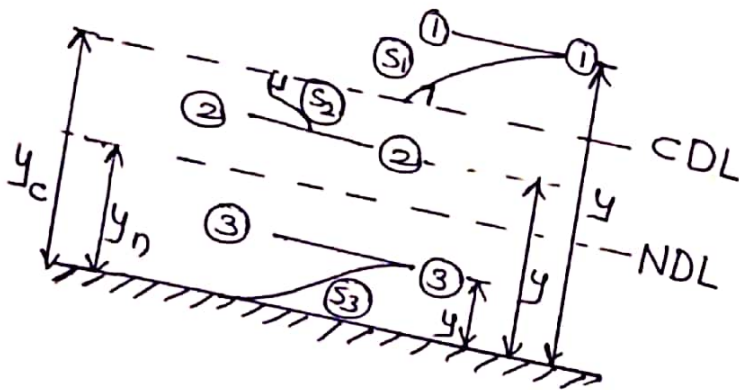
at section ②-②
(M₂ flow profile)

$$y_n > y > y_c$$

At section ③-③
(M₃ flow profile)

$$y_n > y_c > y$$

② Steep Slope:- $y_c > y_n$ - super-critical flow.



At section ①-①
[S₁ - flow profile]

$$y > y_c > y_n$$

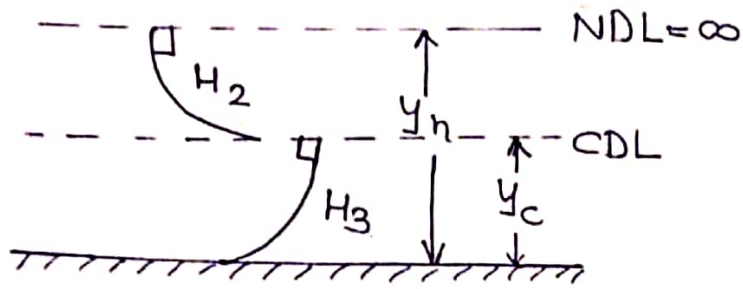
At section ②-②
[S₂ - flow profile]

$$y_c > y > y_n$$

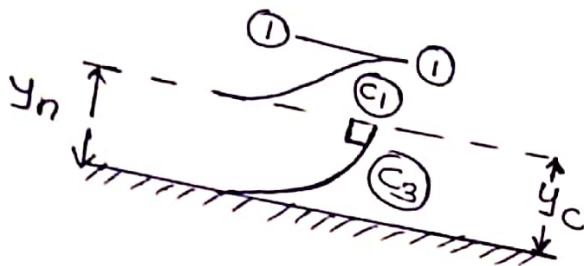
At section ③-③
[S₃ - flow profile]

$$y_c > y_n > y$$

③ Horizontal Slope :- (H-profile) ($y_n > y_c$) - Subcritical flow

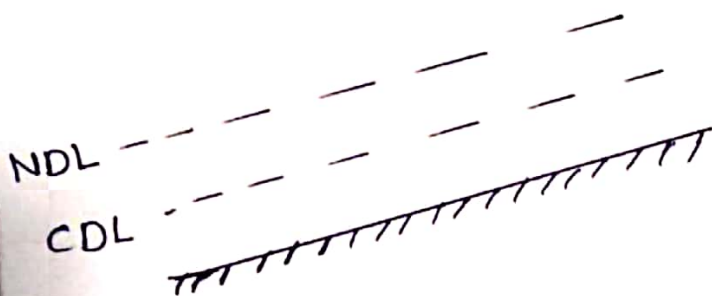


④ Critical Slope (C-profile) ($y_n = y_c$) - critical flow.



⑤ Adverse slope [A-profile] - Flow is subcritical, only direction is opposite.

A_2 & A_3 profile exist only.



Control Section :-

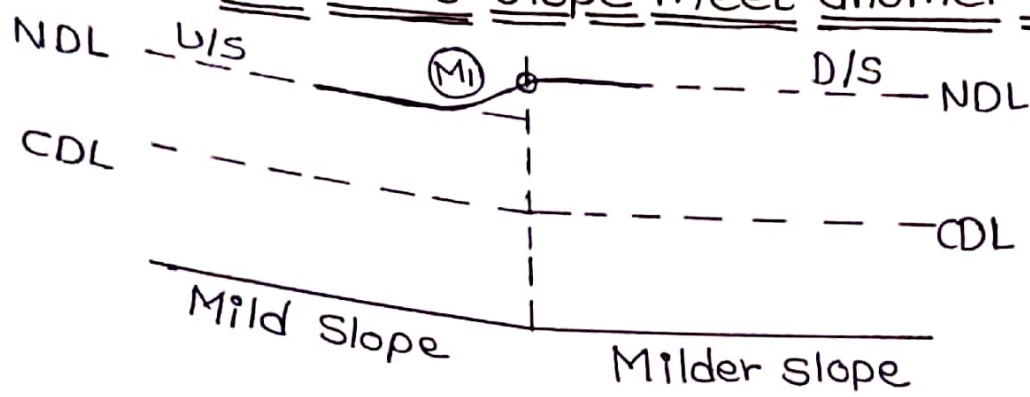
→ A control section is defined as a section in which fixed relationship exist between depth of flow & Discharge.

For example :- Dam, Wier, barrage, sluice gate, head regulator etc.

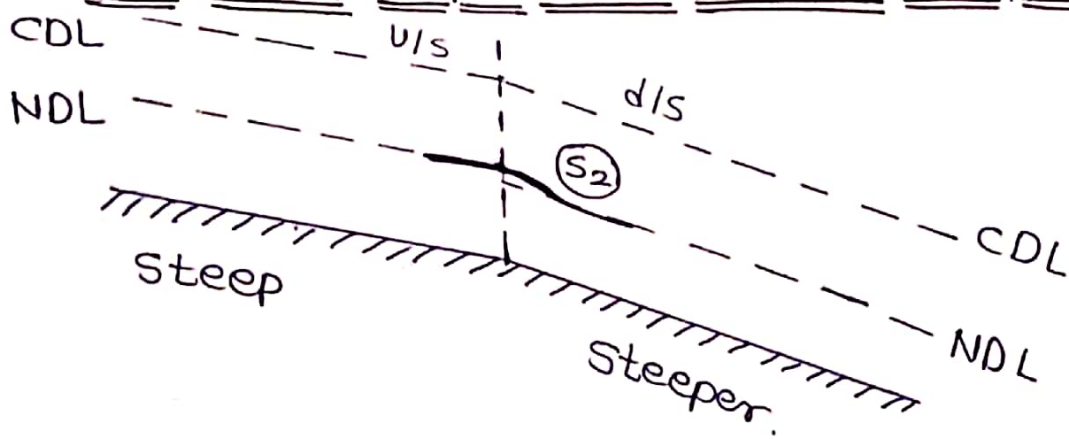
→ The critical depth is also a control point. because they indicate Sub-critical or Supercritical or critical & they also indicate changes in flow i.e. Supercritical to Subcritical or Subcritical to Supercritical.

- * Each GVF profile atleast one control section.
- * In subcritical flow having control section updownstream & in supercritical flow having control section upstream.
- * If Subcritical flow meet supercritical flow so control section having CDL.
- * If Supercritical flow meet subcritical flow, so hydraulic jump is formed.

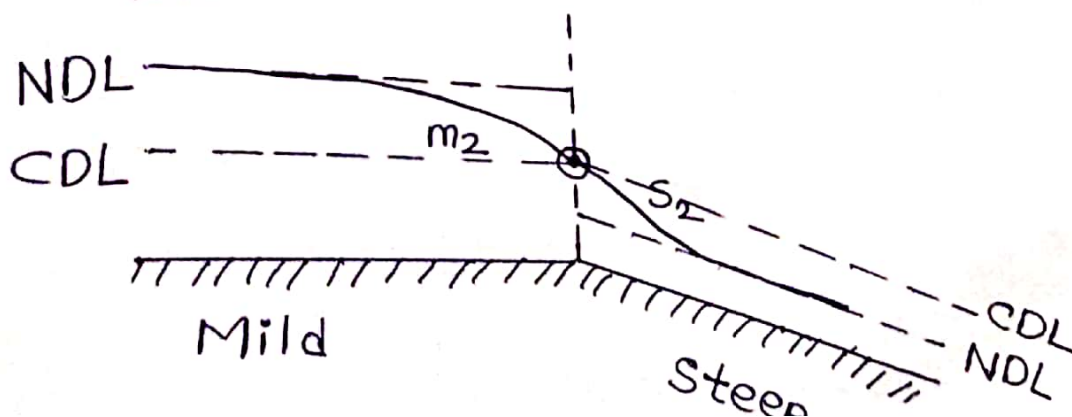
Case ① If Mild Slope meet another Milder Slope:-



Case ② If steep slope meet another steeper slope:-



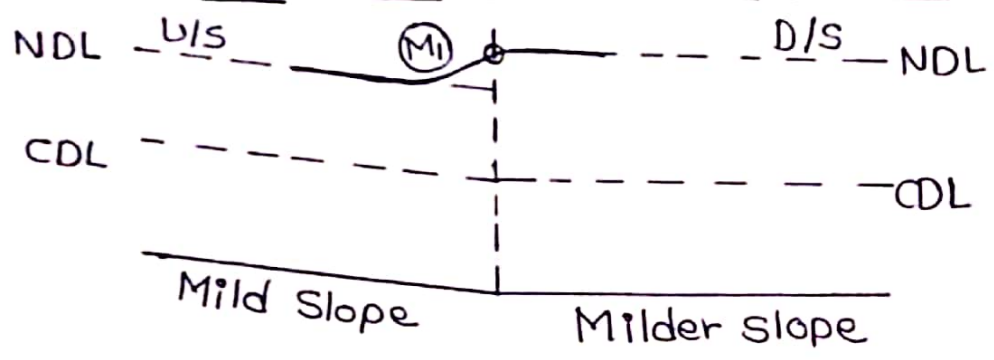
Case ③ If mild slope meet steep slope:-



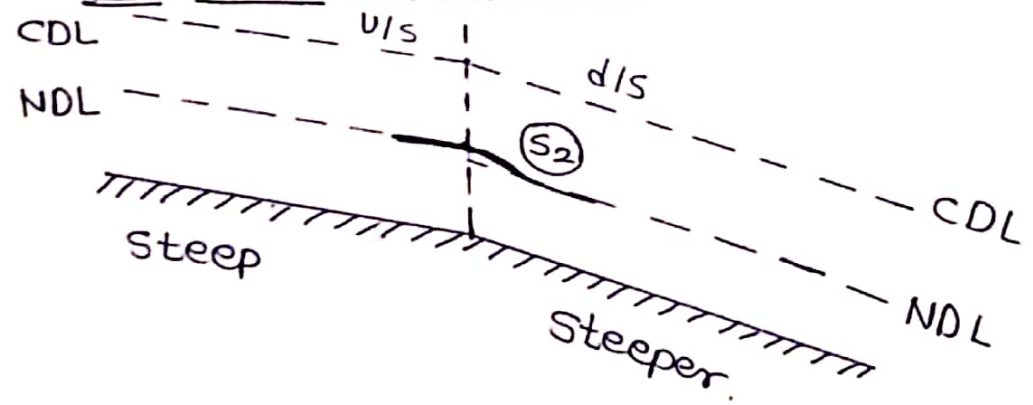
M₂S₂ profile

- * Each GVF profile atleast one control section.
- * In subcritical flow having control section up stream & in supercritical flow having control section upstream.
- * If Subcritical flow meet supercritical flow so control section having CDL.
- * If Supercritical flow meet subcritical flow, so hydraulic jump is formed.

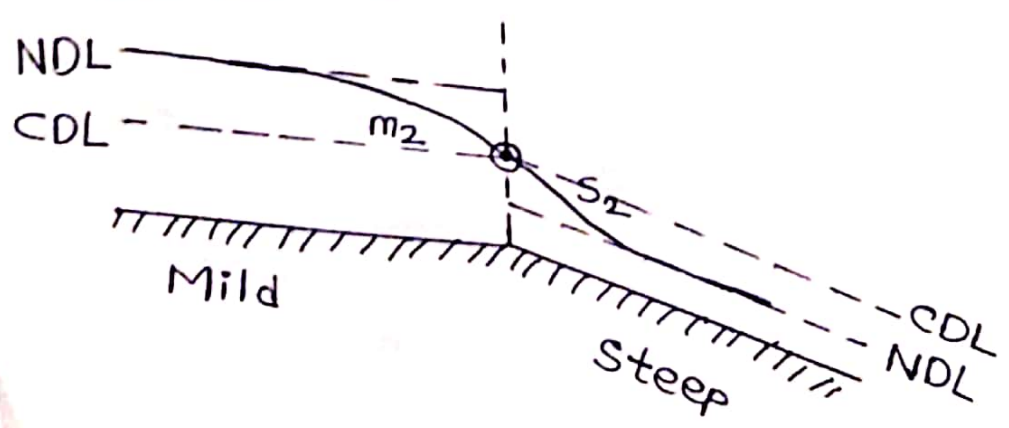
Case ① If Mild Slope meet another Milder Slope:-



case ② If steep slope meet another steeper slope:-



case ③ If mild slope meet steep slope:-

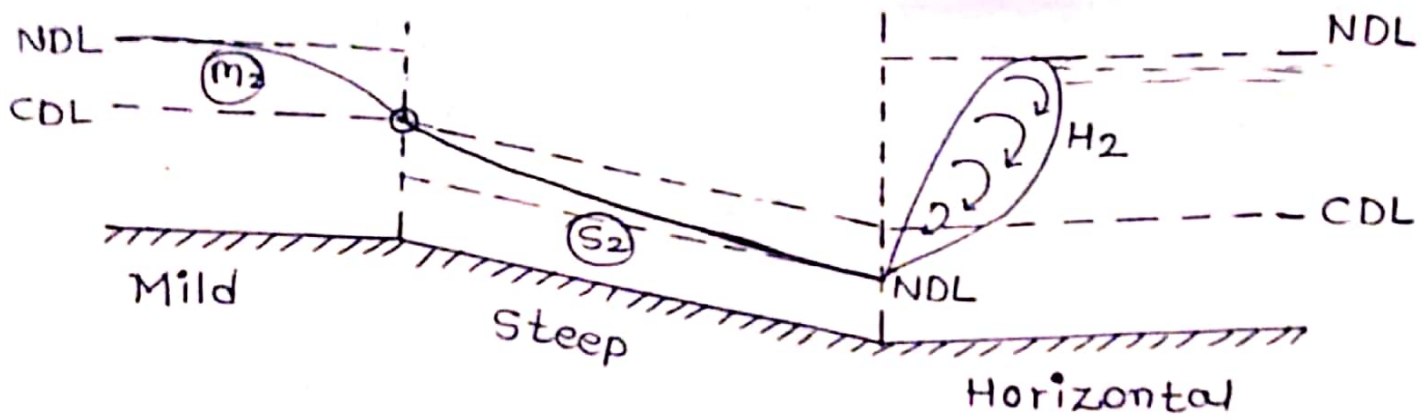


M₂S₂ profile

④ Mild - Steep - Horizontal

M₂S₂H₂ Profile

(32)



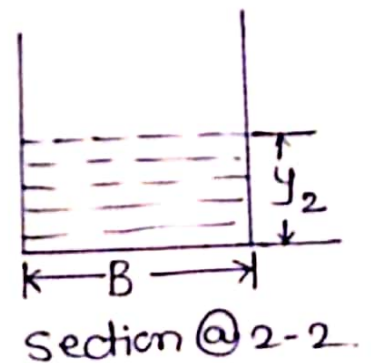
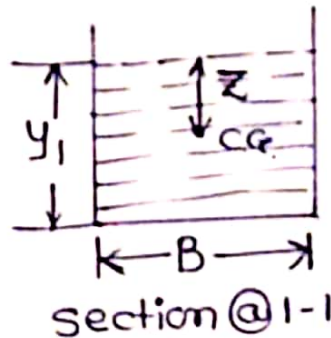
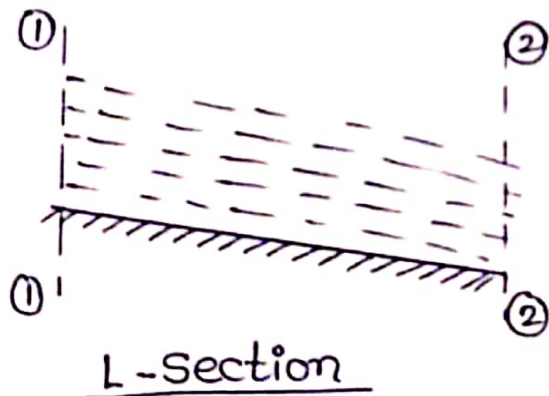
Rapidly Varied Non-Uniform Flow :-

* The flow in open channel is called rapidly varied flow if the flow depth change rapidly over a shorter distance.

e.g. Hydraulic Jump, Surge.

Assumptions:- ① Always be consider channel is Rectangular.

Specific force -



Specific force -

$$F = A \cdot \bar{z} + \frac{Q^2}{g \cdot B \cdot y}$$

at section ①-①:-

$$F_1 = A_1 \bar{z}_1 + \frac{Q^2}{g B y_1}$$

A = wetted Area.

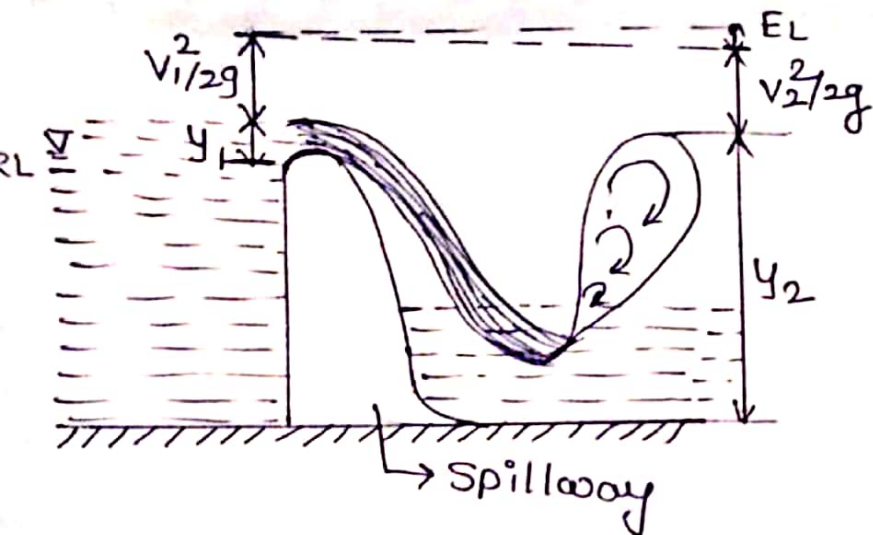
$\bar{z}$ = Distance betⁿ Top Surface of water to the C.G. of wetted area.

at section ②-②:-

$$F_2 = A_2 \bar{z}_2 + \frac{Q^2}{g B y_2}$$

Hydraulic Jump :-

(33)



y_1 & y_2 are sequent depth in which sp. force is same.

→ If higher velocity & Lower depth flow profile meet another higher depth & Lower velocity flow profile, so hydraulic jump is formed.

OR

→ If Super critical flow meet subcritical flow, so hydraulic jump is formed.

$$F_1 = F_2$$
$$A_1 \bar{x} + \frac{Q^2}{g \cdot B y_1} = A_2 \bar{x} + \frac{Q^2}{g \cdot B y_2}$$

$$\frac{y_2}{y_1} = \frac{1}{2} \left[1 + \sqrt{1 + 8 Fr_1^2} \right]$$

$$\frac{y_1}{y_2} = \frac{1}{2} \left[-1 + \sqrt{1 + 8 Fr_2^2} \right]$$

$$Fr_1 = \frac{V_1}{\sqrt{g y_1}}$$

$$Fr_2 = \frac{V_2}{\sqrt{g y_2}}$$

y_1 → Sequent depth pre jump.

y_2 → Sequent depth post jump.

Energy Loss (E_L):-

$$E_L = E_1 - E_2$$

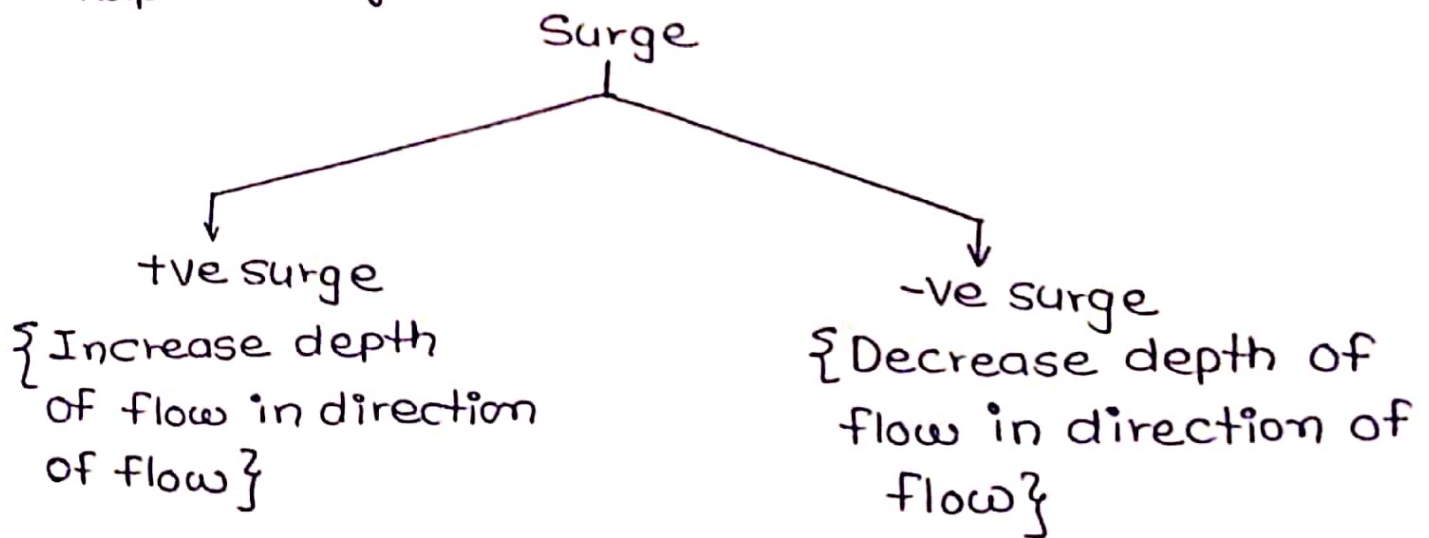
$$E_L = \frac{(y_1 - y_2)^3}{4y_1 y_2} \text{ (m)}$$

Application Of Hydraulic Jump:-

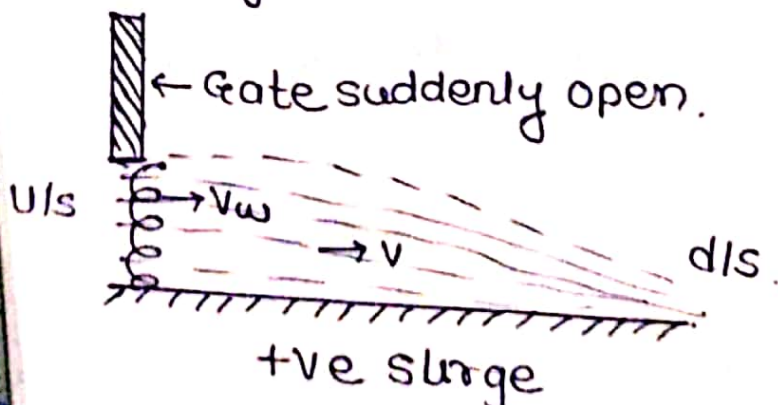
- ① It is used to dissipate the energy below spillway.
- ② It is used for mixing of chemical in water treatment plant.
- ③ Desalination of sea water.
- ④ It can also be used to increase the water load on apron.

Surges in Open channel flow:-

→ Surge is moving wave which produced due to Rapid change of depth of flow.



case ① When Gate provided at head of channel is Suddenly opened.



V_w = velocity of moving wave.
 V = velocity of flow.

Celerity :- $(c)(m/s)$ -

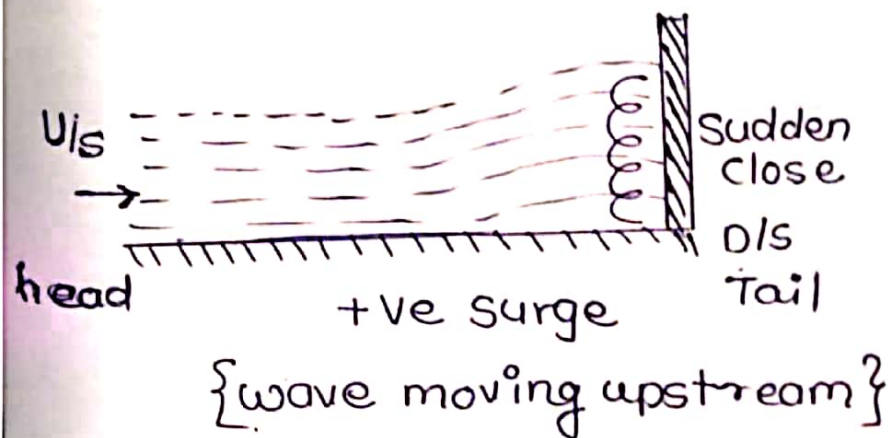
→ celerity is a velocity of moving wave w.r.to velocity of flow.

$$C = V - V_w \quad \{ \text{Wave moving downstream} \}$$

$$C = \sqrt{\frac{g y_2 (y_1 + y_2)}{2 y_1}}$$

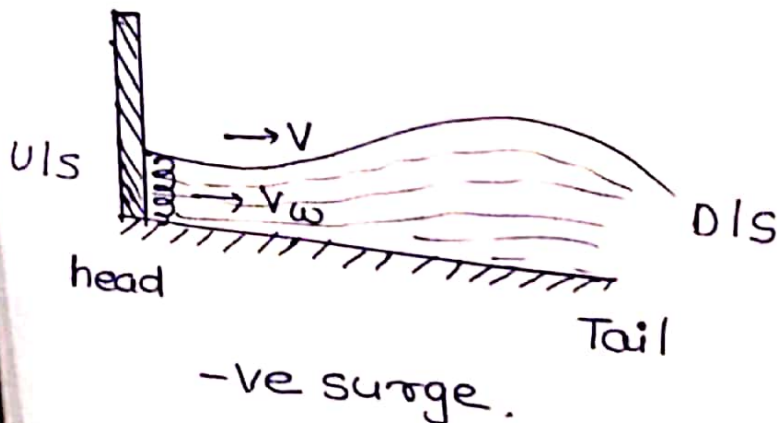
$$\text{when } y_1 = y_2 \quad C = \sqrt{g y}$$

Case 2 - when gate provided at Tail of channel is Sudden closed -



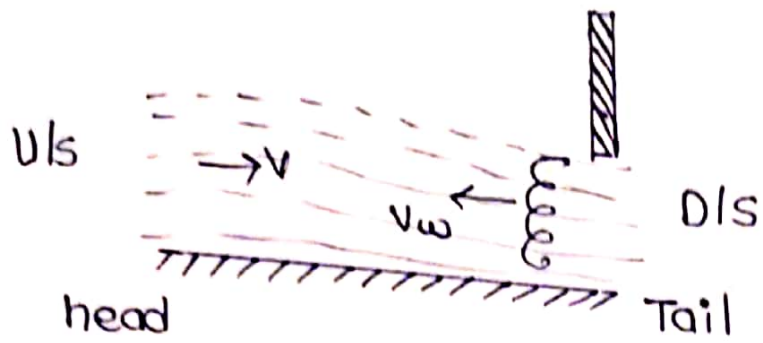
$$\therefore C = V + V_w$$

Case 3 - when gate provided at head of channel is Suddenly closed.



$$C = V - V_w$$

Case 4:- when gate provided at Tail of the channel is suddenly open.



$$C = v + v_w$$

{wave moving U/s side}